

Studio di funzione - Esercizio

domenica 24 novembre 2024 19:15

$$4) y = \frac{x^2 - 16}{25 - x^2}$$

1. DOMINIO

$$25 - x^2 \neq 0 \rightarrow x^2 \neq 25 \rightarrow x \neq \pm 5$$

$$D: \mathbb{R} - \{-5, +5\}$$

$$D: (-\infty, -5) \cup (-5, 5) \cup (5, +\infty)$$

2. SIMMETRIE

$$f(-x) = f(x) \quad \text{PARI}$$

$$\frac{(-x^2) - 16}{25 - (-x)^2} = \frac{x^2 - 16}{25 - x^2} \quad \text{PARI!}$$

NON È DISPARI

3. INTERSEZIONE CON GLI ASSI

$$\begin{cases} y = \frac{x^2 - 16}{25 - x^2} \\ x = 0 \end{cases} \rightarrow \begin{cases} y = \frac{-16}{25} \\ x = 0 \end{cases} \rightarrow \begin{cases} y = -\frac{16}{25} \\ x = 0 \end{cases} \quad (0, -\frac{16}{25})$$

$$\begin{cases} \frac{x^2 - 16}{25 - x^2} = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} x = \pm 4 \\ y = 0 \end{cases} \rightarrow \begin{pmatrix} -4, 0 \\ 4, 0 \end{pmatrix}$$

4. SEGNO

$$x < -4 \vee x > 4$$

$$\frac{x^2 - 16}{25 - x^2} \geq 0$$

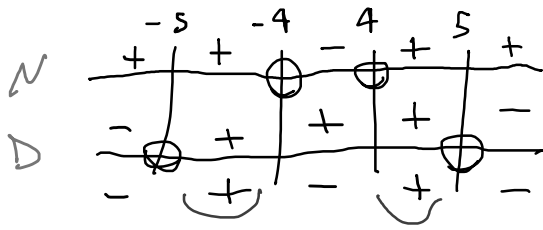
$$N \quad x > \pm 4$$

$$\frac{x^2 - 16}{25 - x^2} \geq 0$$

-5

$$D \quad -x^2 > -25 \rightarrow x < \pm 5$$

-5



$$S = (-5, -4) \cup (4, 5)$$

5. ASINTOTI

$$D: (-\infty, -5) \cup (-5, 5) \cup (5, +\infty)$$

$$\bullet \lim_{x \rightarrow -\infty} \frac{x^2 - 16}{25 - x^2} = \lim_{x \rightarrow -\infty} \frac{x^2(1 - \frac{16}{x^2})}{x^2(25/x^2 - 1)} = \frac{1}{-1} = -1$$

$$\bullet \lim_{x \rightarrow +\infty} \dots = \frac{1}{-1} = -1$$

$$\bullet \lim_{x \rightarrow -5^-} \frac{x^2 - 16}{25 - x^2} = \frac{25 - 16}{25 - 25^-} = \frac{9}{0^-} = -\infty$$

$\rightarrow -25.01$

$$\bullet \lim_{x \rightarrow -5^+} \frac{x^2 - 16}{25 - x^2} = \frac{25 - 16}{25 - 25^+} = \frac{9}{0^+} = +\infty$$

$\rightarrow -24.99$

$$\bullet \lim_{x \rightarrow 5^-} \frac{x^2 - 16}{25 - x^2} = \frac{25 - 16}{25 - 25^-} = \frac{9}{0^+} = +\infty$$

$$\bullet \lim_{x \rightarrow 5^+} \frac{x^2 - 16}{25 - x^2} = \frac{25 - 16}{25 - 25^+} = \frac{9}{0^-} = -\infty$$

ASII
VE

6. GRAFICO

